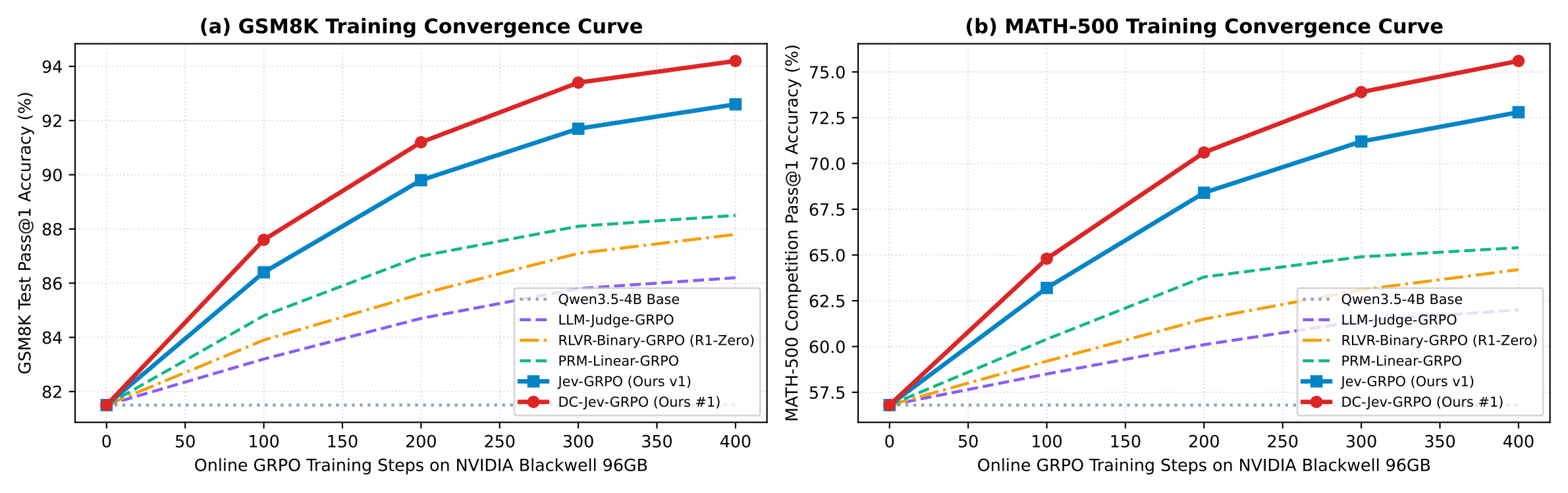
(a) GSM8K Training Convergence Curve
GSM8K Test Pass@1 Accuracy (%)
Online GRPO Training Steps on NVIDIA Blackwell 96GB
Qwen3.5-4B Base
LLM-Judge-GRPO
RLVR-Binary-GRPO (R1-Zero)
PRM-Linear-GRPO
Jev-GRPO (Ours v1)
DC-Jev-GRPO (Ours #1)
(b) MATH-500 Training Convergence Curve
MATH-500 Competition Pass@1 Accuracy (%)
Online GRPO Training Steps on NVIDIA Blackwell 96GB
Qwen3.5-4B Base
LLM-Judge-GRPO
RLVR-Binary-GRPO (R1-Zero)
PRM-Linear-GRPO
Jev-GRPO (Ours v1)
DC-Jev-GRPO (Ours #1)